Computer Networks & Communications: Architecture

Chapter 1: Network Fundamentals and Topologies

Computer networks interconnect autonomous computational devices for resource sharing and communication. Topologies describe how nodes are organized physically and logically: Star (central hub/switch), Bus (shared backbone cable), Ring (token circulation), Mesh (redundant point-to-point connections), and Hybrid. Packet switching divides data into discrete packets routed independently across routers using store-and-forward mechanisms.

Chapter 2: The OSI 7-Layer Reference Model

Detailed Architecture of the OSI Model

The Open Systems Interconnection (OSI) reference model is a conceptual framework standardized by ISO that partitions network communication into seven hierarchical layers: 1. Physical Layer (Layer 1): Transmits raw unstructured bit streams over physical transmission media (cables, optical fibers, radio frequencies). 2. Data Link Layer (Layer 2): Ensures node-to-node frame delivery, physical MAC addressing, framing, and error detection (Ethernet, Wi-Fi). 3. Network Layer (Layer 3): Responsible for logical addressing (IP addresses), packet routing across multi-hop networks, and fragmentation. 4. Transport Layer (Layer 4): Provides end-to-end process-to-process message transfer, port addressing, flow control, and reliability (TCP/UDP). 5. Session Layer (Layer 5): Establishes, manages, synchronizes, and terminates communication sessions between cooperating applications. 6. Presentation Layer (Layer 6): Translates, encrypts, and compresses data formats between the network and application (e.g., TLS, ASCII, JPEG). 7. Application Layer (Layer 7): Provides network interfaces and protocols directly to end-user software applications (HTTP, DNS, SMTP, FTP).

Chapter 3: Transport Layer Protocols: TCP vs UDP

Detailed Comparison Between TCP and UDP

The Transmission Control Protocol (TCP) and User Datagram Protocol (UDP) are the two core protocols of the Transport Layer. Differences between TCP and UDP: - Connection State: TCP is connection-oriented, requiring a 3-way handshake (SYN, SYN-ACK, ACK) prior to data transfer. UDP is connectionless and sends datagrams without establishing a handshake. - Reliability: TCP guarantees delivery through sequence numbers, acknowledgments (ACK), and retransmissions of lost packets. UDP provides best-effort delivery with no acknowledgment or automatic retransmission. - Ordering: TCP guarantees in-order packet delivery using sequence numbers. UDP packets may arrive out of order. - Overhead and Speed: TCP has a larger 20-byte header, congestion control, and flow control mechanisms, making it slower with higher overhead. UDP has a lightweight 8-byte header and low latency. - Use Cases: TCP is used where data accuracy is critical (HTTP/HTTPS, File Transfer FTP, SSH, email). UDP is used for real-time applications where low latency outweighs packet loss (VoIP, video streaming, DNS queries, online multiplayer games).